

**AMARAVATI QUANTUM VALLEY · GOVERNMENT OF ANDHRA
PRADESH**

Quantum & AI Innovation Centre (QAIC)

100 High-Impact Quantum Use Cases

A Strategic, Sector-Wise Compendium for the Quantum Innovation Hub

*Healthcare · Pharmaceuticals · Finance · Cybersecurity · Agriculture · Logistics · Energy ·
Mobility · Manufacturing · Climate · Governance · Telecom · Defence · Education · Space*

Aligned to QAIC development completion — October 2026

Amaravati Quantum Valley Initiative
May 2026

DST-Awarded Quantum Algorithm Proposals — Andhra Pradesh

Under India's National Quantum Mission (NQM), the Department of Science & Technology (DST) has awarded four quantum-algorithm research proposals carrying direct Andhra Pradesh institutional participation. These four awards form the scientific backbone of the QAIC's algorithm-development agenda: every sector use case in this compendium is designed to build on, feed into, or complement the methods being advanced through them. They are presented first, and prominently, because they represent committed national funding and concrete research capacity already secured for the State.

Source: Official DST / NQM list, “List of Quantum Algorithm Proposals – Andhra Pradesh” (proposal records 136428, 134836, 136815, 136876).

01

Quantum Machine Learning, Reservoir Computing and PDE Solvers

Lead Investigator: Prof. M. S. Santhanam — IISER Pune

Member Pls: Dr. Reetesh Kumar Gangwar (IIT Tirupati); Dr. S. Aravinda (IIT Tirupati)

Proposal No.: 136428 (TPN)

Anchors in this compendium: Quantum ML and differential-equation solvers — cross-cutting foundation for healthcare, energy, climate and engineering use cases.

02

Towards Quantum Advantage: Design and Development of Quantum Algorithms, Codes and Protocols

Lead Investigator: Dr. Prabha Mandayam — IIT Madras

Member Pls: Dr. S. Aravinda (IIT Tirupati)

Proposal No.: 134836

Anchors in this compendium: Foundational quantum algorithms, error-correcting codes and protocols — underpins the cybersecurity and quantum-communication use cases.

03

Quantum Annealing and Optimization: Mobility, Multi-modal Logistics, Railway Resource Scheduling and Related Problems

Lead Investigator: Dr. Atanu Rajak — IIT Hyderabad

Member Pls: Dr. Sambuddha Sanyal (IISER Tirupati)

Proposal No.: 136815

Anchors in this compendium: Optimisation for mobility and multi-modal logistics — directly anchors the Logistics, Mobility and Governance use cases.

04

QUEST — Quantum-Enhanced WRF Model for Urban Climate System Modelling**Lead Investigator:** Prof. P. Arumugam — IIT Roorkee**Member PIs:** Prof. Valli Kumari (Andhra University)**Proposal No.:** 136876**Anchors in this compendium:** *Quantum-enhanced weather and climate modelling — anchors the Climate, Energy and disaster-resilience use cases.*

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Executive Summary

This compendium presents 100 high-impact, real-world use cases for the Quantum & AI Innovation Centre (QAIC), structured to advance Andhra Pradesh's ambition of becoming a global quantum innovation hub anchored by the Amaravati Quantum Valley (AQV). It translates the current state of practice in quantum computing, quantum AI and quantum communication into concrete, sector-specific opportunities mapped to India's National Quantum Mission (NQM) and to the State's economic and governance priorities.

The number is deliberate. AQV's stated objective is to deploy an IBM Quantum System Two (133-qubit Heron processor) and to build the capability to develop and test 100 quantum algorithms as part of its readiness programme. This document supplies that portfolio: 100 use cases across 15 sectors, each presented in a uniform, decision-ready format and sequenced so that scoping, algorithm development and proof-of-concept work can be completed by the QAIC development milestone of October 2026.

Four of these directions already carry committed national funding: the DST-awarded quantum-algorithm proposals presented at the front of this document, which involve Andhra Pradesh institutions and anchor the wider portfolio in quantum machine learning, foundational algorithms and codes, optimisation for mobility and logistics, and quantum-enhanced climate modelling.

At a glance

Dimension	Summary
Total use cases	100
Sectors covered	15
Quantum modalities	Quantum Computing · Quantum AI / ML · Quantum Communication (with Quantum Sensing)
DST-awarded algorithm proposals (AP)	4 — highlighted at the start of this document
Presentation format	Sector-wise comparison tables (problem · quantum approach · impact · applicability · partners)
Development milestone	QAIC development completion — October 2026
Delivery network	QAIC statewide hub-and-spoke node network across the Amaravati, Visakhapatnam and Rayalaseema regions

1. Healthcare & Life Sciences

8 use cases

Healthcare is among the most computationally demanding and socially consequential domains for quantum technology. For Andhra Pradesh, where public-health delivery, diagnostics access and the Arogyasri ecosystem touch tens of millions of citizens, quantum machine learning for diagnosis and quantum optimisation for hospital operations offer measurable value, while quantum simulation underpins breakthroughs in genomics and molecular medicine.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-001	Early cancer detection from liquid biopsies Cancers detected late carry far worse outcomes; current screening is invasive, costly and limited in sensitivity for early-stage disease.	Quantum AI / ML Quantum support vector machines (QSVM) and hybrid quantum-classical classifiers analyse cancer-related DNA and cells from blood samples, capturing complex feature relationships that improve specificity and sensitivity over classical classifiers.	Earlier, cheaper, minimally invasive detection enabling better risk stratification and treatment selection; demonstrated f1-scores above 0.9 for several cancer pairs in NQCC trials. Applies to: Health, Medical & Family Welfare Dept; oncology hospitals; Arogyasri network; diagnostic chains.
UC-002	Multi-omics clustering for precision oncology Integrating genomics, proteomics and metabolomics to characterise tumours is computationally expensive and scales poorly with data complexity.	Quantum AI / ML Quantum kernel methods and unsupervised quantum clustering capture multivariate omic patterns and trans-omics pathway activity that classical clustering misses, improving scalability and accuracy.	More meaningful patient sub-typing and pathway insight, paving the way toward personalised cancer treatment regimens. Applies to: Tertiary cancer centres; genomics labs; research hospitals; biobanks.
UC-003	Medical imaging enhancement and radiology triage Radiology backlogs and inter-reader variability delay diagnosis; classical deep-learning models are data- and compute-hungry.	Quantum AI / ML Quantum and quantum-inspired feature maps compress and represent high-dimensional imaging data efficiently, enabling faster triage models and anomaly flagging on limited annotated datasets.	Reduced diagnostic turnaround and improved prioritisation of urgent cases in resource-constrained district hospitals. Applies to: Public health system; teleradiology; primary health centres.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-004	Hospital operations optimisation (rostering, theatres, scanners) Staff rostering, operating-theatre planning and diagnostic-scanner allocation are large combinatorial problems where poor scheduling wastes capacity and lengthens waits.	Quantum Computing Quantum and hybrid optimisation (QAOA, annealing, VQE-derived methods) encode these as constrained optimisation problems; tuned algorithms have shown up to ~10,000x fewer circuit evaluations than naive implementations.	Higher utilisation of scarce clinical assets, shorter elective waits and lower operating cost per patient. Applies to: Hospital administration; teaching hospitals; Arogyasri empanelled facilities.
UC-005	Accelerated genomic sequence analysis and variant calling Population-scale genomics generates data volumes that strain classical alignment and variant-calling pipelines.	Quantum AI / ML · Quantum Computing Novel classical-to-quantum encodings (e.g. reverse-complement embedding that exploits DNA symmetry) and quantum search subroutines accelerate pattern matching and feature extraction on genomic data.	Faster, cheaper population genomics supporting hereditary-disease screening and pharmacogenomics for AP citizens. Applies to: Genome programmes; public-health genomics; research institutes.
UC-006	Protein structure and binding-site prediction Accurately predicting protein folding and ligand binding for difficult targets exceeds the reach of classical approximation at biologically relevant scale.	Quantum Computing Fault-tolerant quantum simulation of molecular electronic structure and quantum-enhanced sampling improve the accuracy of conformational and binding-energy prediction.	Higher-confidence target identification, reducing failed downstream wet-lab experiments and accelerating therapeutic pipelines. Applies to: Pharma R&D; structural biology; academic medical research.
UC-007	Epidemic modelling and contact-network optimisation Forecasting outbreak spread and optimising intervention placement across complex mobility and contact networks is computationally hard.	Quantum Computing · Quantum AI / ML Quantum optimisation over graph sub-problems and QML on mobility data improve scenario modelling, testing-site placement and resource pre-positioning.	Faster, better-targeted public-health responses and more efficient use of vaccines, testing and isolation capacity. Applies to: Public Health Dept; disaster management; municipal administration.
UC-008	Privacy-preserving federated health AI across hospitals	Quantum AI / ML · Quantum Communication	Regulation-compliant (DPDP Act/HIPAA-style) collaborative AI that improves models while protecting confidentiality.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
	Sensitive patient data cannot be centralised, yet siloed datasets limit model quality and equity across facilities.	Federated quantum machine learning combined with fully homomorphic encryption and quantum-secured channels allows model training across distributed hospital data without exposing records.	Applies to: State health data infrastructure; hospital consortia; insurance.

2. Pharmaceuticals & Drug Discovery

7 use cases

Quantum simulation directly attacks the central bottleneck of pharmaceutical R&D: accurately modelling molecular and quantum-chemical behaviour. With India a global generics and bulk-drug power and AP hosting significant pharma and bulk-drug capacity, quantum-accelerated discovery and supply-chain optimisation can compress timelines, reduce attrition and create high-value intellectual property.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-009	Enzyme-targeted drug discovery for difficult targets Metalloenzymes, transition-state systems and covalent inhibitors involve strong electron correlations that classical methods describe poorly, leaving large drug-target classes effectively undruggable.	Quantum Computing Quantum algorithms (VQE with automated active-space selection via QC-DMRG, plus quantum-dynamics methods) model these correlated systems and bond formation more faithfully.	Access to previously intractable targets and new active-space selection methods that deliver value even within classical workflows today. Applies to: Pharma R&D; computational chemistry; CROs.
UC-010	Quantum simulation for novel small-molecule candidates Ground-state and reaction-energy calculations for candidate molecules require approximations that limit accuracy and confidence.	Quantum Computing Resource-estimated fault-tolerant simulation and improved quantum embedding compute molecular properties from first principles with reduced qubit requirements.	Higher-fidelity candidate ranking, lowering costly experimental attrition. Applies to: Drug discovery; agrochemicals; specialty chemicals.
UC-011	Generative quantum chemistry for de novo molecule design Exploring vast chemical space for molecules with desired properties is slow and biased by existing libraries.	Quantum AI / ML Quantum generative models and quantum-enhanced sampling propose novel candidate structures conditioned on target properties.	Broader, faster exploration of chemical space and stronger composition-of-matter IP. Applies to: Pharma & biotech R&D; materials.
UC-012	Clinical trial design and patient-cohort optimisation Site selection, cohort stratification and adaptive trial design are complex optimisation and	Quantum Computing · Quantum AI / ML Quantum optimisation balances multi-constraint site/cohort selection while QML predicts recruitment and dropout, improving statistical power.	Shorter, cheaper, more representative trials with higher completion rates. Applies to: Pharma; CROs; medical colleges running trials.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
	prediction problems that affect cost and success rates.		
UC-013	Vaccine antigen design and immunogenicity prediction Designing stable, immunogenic antigens involves modelling complex protein-immune interactions beyond classical reach.	Quantum Computing · Quantum AI / ML Quantum molecular simulation plus QML on immunological datasets improve antigen stability and immune-response prediction.	Faster, more rational vaccine design and pandemic preparedness capability. Applies to: Vaccine manufacturers; public health; biotech.
UC-014	Pharma supply chain and cold-chain optimisation Distributing temperature-sensitive drugs and vaccines across India's geography with minimal wastage is a hard logistics problem.	Quantum Computing Quantum and quantum-inspired optimisation route shipments, place cold-storage and schedule replenishment under reliability and temperature constraints.	Reduced stock-outs and spoilage, improved last-mile availability of essential medicines. Applies to: Health supply chain; pharma distribution; PDS-adjacent logistics.
UC-015	ADMET and toxicity prediction with quantum ML Predicting absorption, distribution, metabolism, excretion and toxicity early avoids expensive late-stage failures.	Quantum AI / ML Quantum feature maps capture subtle structure-property relationships, improving ADMET classifier performance on limited data.	Earlier de-risking of candidates and reduced animal/clinical attrition. Applies to: Pharma R&D; toxicology; regulatory science.

3. Financial Services

8 use cases

Finance combines high-value optimisation, risk and machine-learning problems with stringent regulation, making it one of the fastest-adopting quantum sectors. India's digital-payments scale (UPI), expanding capital markets and fintech depth give AP an opportunity to host quantum-for-finance R&D anchored by banks, payment firms and the IFSC ecosystem.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-016	Credit-card fraud detection Fraud is rare relative to legitimate transactions, creating severe data imbalance that classical models handle imperfectly, causing false positives and missed fraud.	Quantum AI / ML Quantum restricted Boltzmann machines and quantum energy-based models separate complex transaction patterns, achieving competitive precision/recall on imbalanced data.	Fewer false flags and missed frauds; demonstrated near-zero false negatives in NQCC trials. Applies to: Banks; card networks; payment processors.
UC-017	Anti-money-laundering detection AML demands detecting subtle, evolving patterns across huge transaction graphs while minimising costly false alerts.	Quantum AI / ML Quantum unsupervised learning and quantum-inspired tensor-network models plus quantum feature bundling pre-process features to improve detection at reduced computational cost.	Higher detection on high-value transactions with fewer false positives; deployable today via quantum-inspired models. Applies to: Banks; regulators; fintech compliance.
UC-018	Portfolio optimisation Selecting optimal asset allocations under many constraints and large universes is combinatorially explosive.	Quantum Computing Quantum and hybrid optimisation (QAOA, annealing) explore solution spaces for risk-return-constrained portfolio selection more efficiently.	Better risk-adjusted returns and faster rebalancing for asset managers and pension funds. Applies to: Asset management; pension/insurance funds; IFSC entities.
UC-019	Derivative pricing and Monte-Carlo risk Pricing complex derivatives and computing risk metrics rely on Monte-Carlo simulation that is slow to converge.	Quantum Computing Quantum amplitude estimation offers quadratic speed-ups for expectation and risk computations once fault-tolerant hardware matures.	Faster, more frequent risk recalculation enabling tighter capital and hedging decisions. Applies to: Investment banks; treasury; risk management.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-020	Credit risk and default prediction Thin-file and informal-economy borrowers are hard to assess, limiting fair credit access.	Quantum AI / ML Quantum kernel classifiers extract richer signal from alternative and limited data, improving default prediction with fewer features.	Expanded, fairer credit access and lower non-performing-asset risk for lenders. Applies to: Banks; NBFCs; microfinance; financial inclusion programmes.
UC-021	Real-time UPI and digital-payments fraud defence India's enormous real-time payments volume demands sub-second fraud screening at extreme scale and low false-positive rates.	Quantum AI / ML Quantum-inspired anomaly models and quantum feature bundling reduce the dimensionality and training cost of detection models that score transactions in real time.	Protection of citizens against digital fraud at national-scale throughput; directly relevant to UPI and Aadhaar-enabled payments. Applies to: NPCI ecosystem; banks; payment apps; state digital services.
UC-022	Market scenario simulation and algorithmic strategy Generating realistic market scenarios and stress paths for strategy design is data- and compute-intensive.	Quantum Computing · Quantum AI / ML Quantum generative models and quantum sampling produce richer synthetic scenarios and accelerate strategy back-testing.	More robust trading and risk strategies tested against diverse, realistic conditions. Applies to: Hedge funds; prop trading; exchanges.
UC-023	Insurance pricing and catastrophe-risk modelling Pricing policies and modelling correlated catastrophe risks (cyclones, floods) across portfolios is computationally heavy and increasingly climate-sensitive.	Quantum Computing · Quantum AI / ML Quantum optimisation and QML improve correlated-risk estimation and premium optimisation under capital constraints.	More accurate, resilient pricing for climate-exposed regions such as coastal AP. Applies to: Insurers; reinsurers; crop & disaster insurance schemes.

4. Cybersecurity & Quantum-Safe Infrastructure

7 use cases

Quantum computing simultaneously threatens current public-key cryptography and supplies its defence. This is the single most time-critical sector: 'harvest-now, decrypt-later' attacks make migration to post-quantum cryptography and quantum-secured communication a present-day sovereign imperative, directly aligned with the National Quantum Mission's communication hub and India's 1,000 km QKD milestone.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-024	Post-quantum cryptography (PQC) migration for government systems Today's RSA/ECC encryption will be broken by future fault-tolerant quantum computers, exposing decades of sensitive government and citizen data.	Quantum Communication Migrate state systems to NIST-standardised lattice-based PQC algorithms with crypto-agile architectures that allow rapid algorithm replacement.	Lasting confidentiality of citizen, land, health and financial records; sovereign cryptographic resilience. Applies to: All government departments; e-Governance; banking; defence.
UC-025	Quantum Key Distribution for critical infrastructure Power, water, finance and emergency networks need communication that is provably secure against any future computational attack.	Quantum Communication QKD uses quantum physics to distribute encryption keys that cannot be intercepted without detection, deployed over fibre between critical sites.	Physically guaranteed secure links for the most sensitive state infrastructure; aligns with the NQM communication hub's quantum-secure networking goals. Applies to: Power & water utilities; secretariat; data centres; banks.
UC-026	Quantum random number generation for trusted keys Pseudo-random generators used for keys are deterministic and can be a security weakness.	Quantum Communication Quantum RNG derives entropy from quantum measurement, producing truly unpredictable numbers for cryptographic key material.	Stronger keys for every secured system; a cheap, deployable quantum product. Applies to: Banking; identity systems; IoT; defence.
UC-027	Harvest-now-decrypt-later threat audit and crypto-agility Adversaries may already be storing encrypted data to decrypt once quantum	Quantum Communication Systematic cryptographic inventory and risk-scoring tools identify vulnerable systems and prioritise PQC migration with crypto-agile design.	A prioritised, auditable migration roadmap protecting the highest-risk data first. Applies to: Government CISO functions; banks; healthcare; utilities.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
	computers mature; organisations lack visibility into their exposure.		
UC-028	Quantum-safe DLT for land records and supply chains Blockchain-based registries rely on signatures that quantum computers could forge, undermining tamper-evidence.	Quantum Communication Integrate PQC signatures and quantum-secured channels into distributed ledgers for land, supply-chain and certificate systems.	Durable integrity for AP's land-record and provenance systems against future attacks. Applies to: Revenue/land records; agri & pharma provenance; certificates.
UC-029	Quantum-enhanced intrusion and malware anomaly detection Network-threat detection must find rare, evolving malicious patterns in massive traffic with low false-alarm rates.	Quantum AI / ML Quantum and quantum-inspired anomaly detectors model complex traffic distributions to flag novel intrusions.	Earlier detection of advanced threats to state networks and SOC efficiency gains. Applies to: State SOC's; CERT functions; critical infrastructure.
UC-030	State quantum-secure communication backbone The state government lacks a unified, future-proof secure communication fabric across its offices and agencies.	Quantum Communication Build a metro/inter-city QKD-plus-PQC backbone linking the secretariat, data centres and key institutions, leveraging AQV and NQM infrastructure.	A flagship sovereign secure network positioning AP as a quantum-communication leader within India. Applies to: Whole-of-government; defence liaison; finance.

5. Agriculture & Allied Sectors

7 use cases

Agriculture is central to Andhra Pradesh's economy and livelihoods, and is rich in optimisation and prediction problems — from water and fertiliser allocation to pest forecasting and aquaculture. Quantum machine learning and optimisation can sharpen precision-agriculture decisions in the near term, while quantum chemistry promises longer-term breakthroughs in sustainable fertiliser production.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-031	Crop-yield prediction and precision agronomy Predicting yields and tailoring inputs across heterogeneous soils, weather and crops is data-intensive and uncertain.	Quantum AI / ML Quantum ML on satellite, soil and weather data improves yield forecasting and per-plot input recommendations with compact models.	Higher yields and lower input cost for farmers; better procurement and price planning for the state. Applies to: Agriculture Dept; RBKs (Rythu Bharosa Kendras); cooperatives; agri-fintech.
UC-032	Sustainable fertiliser and nitrogen-fixation catalyst design Ammonia production (Haber-Bosch) is energy- and emissions-intensive; nature's nitrogenase enzyme is poorly understood at the quantum level.	Quantum Computing Quantum simulation of nitrogenase and candidate catalysts could unlock low-energy nitrogen fixation, a landmark quantum-chemistry challenge.	Transformational reduction in fertiliser energy use and emissions; major agricultural and climate dividend. Applies to: Fertiliser industry; agri-research; climate.
UC-033	Irrigation and water-resource allocation optimisation Allocating limited water across canals, reservoirs and crops under competing demands and variable inflows is a complex optimisation problem.	Quantum Computing Quantum and quantum-inspired optimisation schedule releases and allocations to maximise productive use under hydrological constraints.	More equitable, efficient water use across AP's Krishna-Godavari command areas; reduced conflict and waste. Applies to: Water Resources Dept; CWC/KGBO coordination; irrigation boards.
UC-034	Pest and crop-disease early warning Late detection of pest and disease outbreaks causes large yield and income losses.	Quantum AI / ML Quantum ML on multispectral/drone imagery and weather data detects early stress signatures and predicts outbreak spread.	Timely, targeted intervention reduces crop loss and pesticide overuse. Applies to: Agriculture & Horticulture Depts; RBKs; insurers.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-035	Agri supply-chain and cold-storage logistics optimisation Post-harvest losses are high due to inefficient routing, storage siting and market matching for perishables.	Quantum Computing Quantum optimisation routes produce, sites cold storage and matches supply to markets to minimise spoilage and cost.	Reduced post-harvest loss and better farmer realisation; improved food security. Applies to: Markets/AGMARKNET; cold-chain operators; FPOs.
UC-036	Soil-carbon and nutrient modelling Quantifying soil carbon and nutrient dynamics for regenerative practices and carbon credits is data-heavy and uncertain.	Quantum AI / ML · Quantum Computing QML on soil-sensor and remote-sensing data improves nutrient and carbon-stock estimation for management and verification.	Enables carbon-farming income and precise nutrient management; supports AP natural-farming initiatives. Applies to: Agriculture Dept; natural-farming programmes; carbon markets.
UC-037	Aquaculture and livestock disease and feed optimisation AP's large shrimp/aquaculture and livestock sectors face disease risk and feed inefficiency that erode margins.	Quantum AI / ML · Quantum Computing QML predicts disease onset from water-quality and sensor data while quantum optimisation tunes feed and stocking schedules.	Lower disease losses, optimised feed cost and higher, more sustainable productivity. Applies to: Fisheries & Animal Husbandry Depts; aqua exporters; cooperatives.

6. Logistics & Transport

7 use cases

Logistics is the canonical quantum-optimisation domain: routing, scheduling, loading and network design are NP-hard problems where even modest improvements yield large savings. With AP's ports, freight corridors and industrial growth, quantum and quantum-inspired optimisation can deliver measurable efficiency and cost reductions.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-038	Last-mile delivery and vehicle routing Routing fleets across dynamic demand, traffic and time windows is combinatorially hard and costly when done sub-optimally.	Quantum Computing Quantum and quantum-inspired solvers tackle vehicle-routing and travelling-salesperson variants at practical scale, often hybridised with classical methods.	Lower fuel, time and emissions for delivery and field-service fleets. Applies to: E-commerce; postal/courier; field services; municipal collection.
UC-039	Port and container-terminal optimisation Berth allocation, yard stacking and crane scheduling at busy ports are interlocking optimisation problems affecting throughput.	Quantum Computing Quantum optimisation coordinates berth, yard and equipment scheduling to raise terminal throughput and cut dwell time.	Higher capacity and competitiveness for AP's ports (Visakhapatnam, Krishnapatnam, Gangavaram, Machilipatnam). Applies to: Ports; maritime board; export logistics.
UC-040	Warehouse slotting and inventory optimisation Placing inventory and planning picking paths to minimise handling is a large combinatorial challenge.	Quantum Computing · Quantum AI / ML Quantum optimisation determines slotting and picking sequences while QML forecasts demand to position stock.	Faster fulfilment and lower warehousing cost. Applies to: Retail & e-commerce DCs; 3PLs; manufacturing stores.
UC-041	Air-cargo container loading and balancing Loading unit-load devices to maximise payload while keeping centre-of-gravity optimal is a hard, safety-critical problem traditionally done manually.	Quantum Computing Quantum annealing and hybrid solvers optimise selection and placement, improving gross/net load factors and balance.	Modelled fuel savings (~thousands of tonnes over long-haul programmes) and faster load planning. Applies to: Airports; cargo airlines; freight forwarders.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-042	Rail freight scheduling and rake optimisation Scheduling freight rakes, paths and yard movements across a congested network is a vast scheduling problem.	Quantum Computing Quantum optimisation improves rake allocation, pathing and yard sequencing under capacity constraints.	Higher freight throughput and reduced detention/idle time. Applies to: Railways; mineral & industrial freight; ports hinterland.
UC-043	Multimodal freight-corridor planning Coordinating road-rail-port-air flows for industrial corridors requires joint optimisation across modes.	Quantum Computing · Quantum AI / ML Hybrid quantum optimisation plus QML demand forecasting plan capacity and routing across modes.	Lower logistics cost as a share of GDP and stronger industrial competitiveness for AP corridors. Applies to: Industrial corridors; infrastructure planning; exporters.
UC-044	Supply-chain network design and resilience Designing resilient supply networks under disruption (pandemics, geopolitics, climate) is a complex multi-objective problem.	Quantum Computing · Quantum AI / ML Quantum optimisation evaluates facility placement, sourcing and inventory buffers for cost-resilience trade-offs.	More robust supply chains that withstand shocks while controlling cost. Applies to: Manufacturing; FMCG; pharma; strategic stockpiles.

7. Energy & Power

7 use cases

The energy transition is computationally bound — from designing better batteries and catalysts to balancing increasingly renewable, distributed grids. AP's strong renewable ambitions and grid-modernisation agenda make energy a priority where quantum optimisation delivers value and quantum simulation enables materials breakthroughs.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-045	Wind-farm siting and turbine-layout optimisation Choosing wind-farm locations and arranging turbines to maximise energy while minimising wake losses is a hard spatial-optimisation problem.	Quantum Computing Quantum optimisation (demonstrated on photonic hardware) addresses deployment modelling and layout, integrating with industry wake models.	Higher energy yield and project viability for onshore/offshore wind. Applies to: NREDCAP; energy developers; utilities.
UC-046	Grid load balancing and unit commitment Scheduling generation and balancing supply-demand on a high-renewable grid is a large mixed-integer optimisation under uncertainty.	Quantum Computing Quantum and hybrid optimisation solve unit-commitment and dispatch to integrate variable renewables more efficiently.	Lower curtailment, cost and emissions; improved grid stability. Applies to: DISCOMs; transmission utility (APTRANSCO); load dispatch.
UC-047	Next-generation battery-materials simulation Designing better cathodes and electrolytes requires modelling complex spin dynamics (e.g. NMR spectra of cobalt/transition-metal materials) beyond classical reach.	Quantum Computing Hybrid quantum workflows simulate NMR spectra and electronic structure from first principles, with resource estimates for fault-tolerant devices.	Faster discovery of safer, higher-capacity batteries supporting EVs and storage. Applies to: Battery & cell manufacturers; EV industry; storage.
UC-048	Solar-farm siting and generation forecasting	Quantum AI / ML · Quantum Computing QML improves irradiance/output forecasting while quantum optimisation aids siting and inverter/array configuration.	Better grid integration and higher returns on AP's large solar pipeline. Applies to: Solar developers; DISCOMs; NREDCAP.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
	Solar output is weather-dependent and siting/forecasting errors raise integration cost.		
UC-049	Green-hydrogen catalyst discovery Efficient, cheap catalysts for electrolysis and fuel cells are needed to make green hydrogen viable.	Quantum Computing Quantum simulation models catalytic reaction pathways and candidate materials with chemical accuracy.	Cheaper green hydrogen, advancing India's and AP's hydrogen-mission goals. Applies to: Hydrogen industry; chemicals; heavy transport.
UC-050	EV charging-network placement and scheduling Placing chargers and scheduling charging to match demand and grid capacity is a coupled siting-and-scheduling problem.	Quantum Computing Quantum optimisation co-optimises charger placement and smart-charging schedules under grid and demand constraints.	Faster EV adoption with minimal grid stress and lower infrastructure cost. Applies to: Urban development; transport; DISCOMs; CPOs.
UC-051	Grid resilience and fault/outage optimisation Restoring power and reconfiguring networks after faults or disasters quickly and safely is a complex optimisation.	Quantum Computing · Quantum AI / ML Quantum optimisation computes restoration sequences and network reconfiguration while QML predicts fault risk.	Faster recovery and fewer/shorter outages, vital for cyclone-prone coastal AP. Applies to: DISCOMs; disaster management; transmission.

8. Mobility & Smart Cities

6 use cases

Urban mobility generates dense, real-time optimisation problems — traffic control, transit scheduling, fleet dispatch — well-suited to quantum and quantum-inspired methods. As Amaravati is built as a new-age capital, embedding quantum-optimised mobility from the start can make it a showcase smart city.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-052	Urban traffic-signal optimisation Coordinating signals across a city to minimise congestion and travel time is a large real-time optimisation problem.	Quantum Computing Quantum optimisation (demonstrated by automakers on annealers) sets adaptive signal timings using live traffic data.	Reduced congestion, travel time and emissions in Amaravati, Vijayawada and Visakhapatnam. Applies to: Urban development; municipal/traffic police; smart-city missions.
UC-053	Public-transport network and bus scheduling Designing routes and crew/vehicle schedules to maximise coverage and reliability at minimum cost is combinatorial.	Quantum Computing Quantum optimisation plans routes, frequencies and crew rosters under demand and resource constraints.	Better service and utilisation for APSRTC and city transit at lower cost. Applies to: Transport Dept; APSRTC; city transit authorities.
UC-054	EV fleet routing and energy-aware dispatch Electric fleets must route considering range, charging windows and grid prices simultaneously.	Quantum Computing Quantum optimisation jointly plans routes, charging and dispatch for electric buses and delivery fleets.	Lower operating cost and emissions while keeping vehicles in service. Applies to: Transit; logistics; municipal fleets.
UC-055	Autonomous-vehicle perception and sensor fusion Fusing multi-sensor data for safe perception under uncertainty is computationally intensive.	Quantum AI / ML Quantum machine learning could enhance pattern recognition and sensor fusion for perception and planning.	Safer, more capable autonomous and assisted-driving systems. Applies to: Automotive R&D; mobility startups; research.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-056	Ride-hailing demand prediction and matching Predicting demand and matching riders to vehicles in real time at scale is a hard prediction-plus-assignment problem.	Quantum AI / ML · Quantum Computing QML forecasts demand hotspots while quantum optimisation solves driver-rider assignment.	Shorter wait times, higher utilisation and lower deadheading. Applies to: Mobility platforms; city transit integration.
UC-057	Roadwork scheduling and maintenance optimisation Scheduling road maintenance across a network to minimise public disruption and cost is a constrained scheduling problem.	Quantum Computing Quantum optimisation allocates work orders to road sections and time windows under multiple constraints, validated on real road-network studies.	Less congestion and public disruption; lower contractor cost and better-maintained roads. Applies to: Roads & Buildings Dept; municipal works; highways.

9. Manufacturing & Industry 4.0

7 use cases

Manufacturing combines scheduling, design and quality problems that map naturally onto quantum optimisation and simulation. As AP pursues industrial growth and Industry 4.0 adoption, quantum-enhanced production planning and materials design can raise productivity and competitiveness.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-058	Job-shop and production scheduling (DFJSSP) Dynamic flexible job-shop scheduling — assigning multi-operation jobs to machines to minimise completion time as jobs arrive — is a hard real-world problem across many industries.	Quantum Computing Quantum annealing and constrained-quadratic-model solvers find optimal/near-optimal schedules, with tailored modelling improving scaling.	Lower makespan, higher machine utilisation and reduced waste across manufacturing. Applies to: Discrete & process manufacturing; foundries; assembly.
UC-059	Alloy design and corrosion modelling Modelling metal corrosion and designing resilient alloys requires accurate quantum-chemical simulation of complex reactions.	Quantum Computing Quantum embedding splits chemical problems across classical and quantum resources to simulate corrosion (e.g. copper-oxygen) and guide alloy design.	More durable materials and better predictive maintenance for aerospace, infrastructure and heavy industry. Applies to: Metals; aerospace; infrastructure; defence.
UC-060	Quality control and defect detection Detecting subtle, rare defects from inspection imagery at line speed is difficult with limited labelled data.	Quantum AI / ML Quantum-enhanced vision models classify defects with strong performance on small datasets.	Lower defect escape rates and inspection cost; higher yield. Applies to: Electronics; automotive; pharma packaging; textiles.
UC-061	Predictive maintenance Predicting equipment failure from sensor streams to avoid downtime is a complex time-series problem.	Quantum AI / ML Quantum time-series and kernel methods model degradation patterns to forecast failures earlier.	Reduced unplanned downtime and maintenance cost. Applies to: Heavy industry; utilities; ports; refineries.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-062	Production planning and demand forecasting Aligning production with uncertain demand across products and constraints is a coupled forecasting-and-optimisation problem.	Quantum Computing · Quantum AI / ML QML forecasts demand and quantum optimisation plans production and procurement under capacity limits.	Lower inventory and stock-outs; improved service levels. Applies to: FMCG; pharma; discrete manufacturing.
UC-063	Computational fluid dynamics for product design High-fidelity CFD for aerodynamics and flow design is expensive, forcing reliance on costly physical testing.	Quantum Computing Quantum CFD algorithms (validated on emulated/early hardware) promise major efficiency gains; ~100 high-fidelity qubits could make some simulations classically intractable.	Faster aerodynamic/thermal design and reduced physical testing for aerospace, automotive and energy. Applies to: Aerospace; automotive; turbomachinery; HVAC.
UC-064	Factory layout and robotics path optimisation Designing factory layouts and coordinating robot/AGV paths to minimise travel and collisions is combinatorial.	Quantum Computing Quantum optimisation plans layouts and multi-robot routing under spatial and timing constraints.	Higher throughput and safer, more efficient automated facilities. Applies to: Smart factories; warehouses; automotive plants.

10. Climate & Environment

6 use cases

Climate action depends on hard computational problems: modelling the atmosphere, designing carbon-capture materials, optimising emissions and forecasting disasters. For coastal, cyclone- and flood-exposed AP, quantum-enhanced disaster forecasting and climate optimisation carry direct resilience value, while quantum simulation underpins decarbonisation materials.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-065	Carbon-capture material (sorbet/MOF) design Designing efficient, cheap CO ₂ -capture materials requires accurate modelling of gas-sorbent interactions.	Quantum Computing Quantum simulation models adsorption chemistry and candidate metal-organic frameworks with chemical accuracy.	Cheaper carbon capture supporting net-zero goals. Applies to: Power & cement industry; climate-tech; chemicals.
UC-066	Enhanced climate and weather modelling Climate and weather models are limited by the cost of resolving fine-scale, nonlinear dynamics.	Quantum Computing · Quantum AI / ML Quantum algorithms for differential equations and QML emulators could accelerate sub-components of climate/weather simulation.	Better seasonal and extreme-event forecasting for agriculture and disaster planning. Applies to: Meteorology; agriculture; disaster management.
UC-067	Flood forecasting and disaster-response sensor placement Placing limited sensors and communication nodes for maximal coverage in disaster zones, and forecasting floods, are coupled hard problems acutely relevant to the Krishna-Godavari basins.	Quantum Computing · Quantum AI / ML Quantum optimisation maximises coverage with minimal nodes (validated on real geographies) and QML improves flood/inflow forecasting.	Faster, better-targeted disaster response and earlier warnings, saving lives and assets. Applies to: SDMA; Water Resources; CWC/KGBO; municipalities.
UC-068	Air-quality monitoring and pollution-source optimisation Identifying pollution sources and siting monitoring/abatement optimally across a city is data- and optimisation-heavy.	Quantum AI / ML · Quantum Computing QML attributes sources from sensor data and quantum optimisation sites monitors and interventions.	Cleaner air and more cost-effective pollution control in urban AP. Applies to: Pollution control boards; urban development; industry.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-069	Carbon-credit and emissions-portfolio optimisation Allocating abatement investments and managing carbon-credit portfolios under budget and target constraints is a complex optimisation.	Quantum Computing Quantum and quantum-inspired optimisation select abatement and offset portfolios to meet targets at least cost.	Cost-efficient decarbonisation pathways for industry and government. Applies to: Industry sustainability; carbon markets; PSUs.
UC-070	Aviation emissions and flight-path optimisation Aviation is a hard-to-abate sector; optimising flight paths and airspace use can cut emissions before new aircraft mature.	Quantum Computing · Quantum AI / ML Graph-based airspace models solved on quantum annealers and emulated circuit machines find shorter, lower-emission paths; hybrid methods address air-traffic bottlenecks.	Reduced aviation fuel burn and emissions; patentable air-traffic optimisation methods. Applies to: Air-traffic management; airlines; airports.

11. Governance & Public Services

7 use cases

Government is itself a vast generator of optimisation, allocation and detection problems — welfare targeting, revenue protection, resource allocation and service delivery. Quantum and quantum-inspired methods can improve the efficiency and integrity of public services, with AP's strong digital-governance base providing a ready platform.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-071	Welfare-scheme targeting and subsidy allocation Targeting benefits to eligible citizens and allocating finite subsidies fairly and efficiently is a large constrained problem prone to exclusion/inclusion errors.	Quantum Computing · Quantum AI / ML QML improves eligibility prediction while quantum optimisation allocates limited resources under equity and budget constraints.	Better-targeted welfare with reduced leakage and exclusion errors. Applies to: Welfare departments; DBT programmes; planning.
UC-072	Tax and GST evasion detection Detecting evasion and fraudulent claims in massive tax/GST datasets with few false positives is challenging.	Quantum AI / ML Quantum anomaly detection and quantum-inspired models flag suspicious filings and circular-trading patterns.	Higher revenue recovery and fairer compliance. Applies to: Commercial Taxes; GST authorities; audit.
UC-073	Public-resource and budget allocation optimisation Allocating budgets, staff and assets across competing priorities and regions is a complex multi-objective optimisation.	Quantum Computing Quantum optimisation evaluates allocation scenarios against outcome and equity objectives.	More effective, transparent use of public funds. Applies to: Finance & Planning Depts; line departments.
UC-074	Citizen-service routing and demand forecasting Routing grievances and forecasting demand at service centres (e.g. one-stop citizen services) affects responsiveness.	Quantum AI / ML · Quantum Computing QML forecasts demand and quantum optimisation routes cases and staffs centres accordingly.	Faster grievance resolution and better-staffed citizen services. Applies to: e-Governance; grievance redressal; service centres.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-075	Election and polling-logistics optimisation Placing polling stations and allocating staff, EVMs and security across districts is a large logistics problem.	Quantum Computing Quantum optimisation allocates resources to maximise accessibility and coverage under constraints.	Smoother, more accessible elections at lower cost. Applies to: Election machinery; district administration.
UC-076	Land-records integrity via quantum-safe ledgers Land-record systems must remain tamper-evident for decades, but current cryptography will become vulnerable.	Quantum Communication Quantum-safe (PQC) signatures and secured channels protect land-record ledgers against future attacks.	Durable, trustworthy land titles reducing disputes and fraud. Applies to: Revenue/Survey & Settlement; registration.
UC-077	Disaster-management resource and evacuation optimisation Pre-positioning relief and planning evacuations under uncertainty across cyclone/flood-prone areas is time-critical and complex.	Quantum Computing Quantum optimisation plans resource pre-positioning, shelter allocation and evacuation routing in real time.	Faster evacuations and better-targeted relief, saving lives in coastal AP. Applies to: SDMA; revenue/district administration; NDRF coordination.

12. Telecommunications

6 use cases

Telecom networks pose continuous optimisation challenges and are also the substrate for quantum communication. With India's massive subscriber base and 5G/6G rollout, AP can pioneer both quantum-optimised network operations and quantum-secured telecom backbones aligned with the NQM communication hub.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-078	5G/6G network optimisation and spectrum allocation Allocating spectrum and configuring dense networks to maximise capacity and minimise interference is combinatorially hard.	Quantum Computing Quantum optimisation assigns spectrum, power and beamforming configurations under interference constraints.	Higher network capacity and quality of service at lower energy cost. Applies to: Telecom operators; TRAI/DoT planning; equipment vendors.
UC-079	Network traffic routing and congestion control Routing traffic dynamically to avoid congestion across complex networks is a hard real-time optimisation.	Quantum Computing · Quantum AI / ML Quantum optimisation and QML predict and route around congestion to balance load.	Lower latency and higher throughput for data and voice. Applies to: Telecom & ISPs; data centres; CDNs.
UC-080	Cell-tower and base-station placement Siting towers for maximal coverage with minimal infrastructure is a coverage-optimisation problem.	Quantum Computing Quantum optimisation (as in disaster-network coverage studies) maximises coverage with the fewest nodes.	Better rural and urban coverage at lower capital cost; supports digital inclusion. Applies to: Telecom operators; BharatNet; rural connectivity.
UC-081	Quantum-secure telecom backbone (QKD over fibre) Telecom backbones carrying sensitive traffic need future-proof security beyond classical encryption.	Quantum Communication QKD integrated into fibre backbones provides physically secure key exchange between exchanges and data centres.	A quantum-secured telecom layer for finance, government and defence traffic. Applies to: Telecom infrastructure; finance; government.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-082	Telecom fraud and SIM-abuse detection Detecting fraud, spam and SIM-box abuse across enormous call/data records with low false positives is difficult.	Quantum AI / ML Quantum and quantum-inspired anomaly detection flag fraudulent usage patterns efficiently.	Reduced fraud losses and better consumer protection. Applies to: Telecom operators; regulators; consumer protection.
UC-083	Quantum-networking and entanglement-distribution node A future quantum internet requires distributing entanglement and quantum memories across nodes — frontier R&D today.	Quantum Communication Establish an AP quantum-networking testbed exploring entanglement distribution, quantum repeaters and memories.	Positions AP at the frontier of the quantum internet and trusted distributed computing. Applies to: Research; defence; future secure computing.

13. Defence & Strategic Applications

6 use cases

Defence applications of quantum technology are among the closest to fruition in sensing and communication, and strategically vital in cryptography. While defence is a national/Union domain, AP's quantum ecosystem can contribute research, talent and dual-use innovation in partnership with national agencies.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-084	Secure military and strategic communications (QKD) Military communications require provably unbreakable confidentiality against current and future adversaries.	Quantum Communication QKD and PQC provide tamper-evident, future-proof secure links for strategic communications.	Sovereign, attack-resistant command and control communications. Applies to: Defence; strategic agencies; secure government.
UC-085	Quantum sensing for GPS-denied navigation Navigation in GPS-denied environments needs drift-free positioning unavailable to classical inertial systems.	Quantum Sensing Quantum inertial sensors (cold-atom accelerometers/gyroscopes) and quantum magnetometers enable precise navigation without satellites.	Resilient navigation for defence, aviation and maritime; aligns with NQM sensing hub. Applies to: Defence; aviation; maritime; survey.
UC-086	Radar and sonar signal-processing enhancement Extracting weak signals from noisy radar/sonar returns at speed is computationally demanding.	Quantum AI / ML · Quantum Computing Quantum-enhanced signal processing and QML improve detection and classification in cluttered environments.	Better situational awareness and threat detection. Applies to: Defence; coastal surveillance; maritime security.
UC-087	Defence logistics and mission planning optimisation Planning logistics, asset allocation and mission routing under many constraints is a hard optimisation.	Quantum Computing Quantum optimisation plans resource allocation, routing and scheduling for complex operations.	Faster, more efficient strategic logistics and planning. Applies to: Defence logistics; strategic planning.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-088	Cryptanalysis-resistant command systems (PQC) Strategic systems must be hardened now against future quantum cryptanalysis.	Quantum Communication Deploy PQC across command-and-control and weapons-platform communications with crypto-agility.	Durable security of strategic systems and stored secrets. Applies to: Defence; strategic enterprises; critical infrastructure.
UC-089	Materials for stealth and high-performance defence components Designing radar-absorbing, high-temperature and lightweight materials needs accurate quantum-scale modelling.	Quantum Computing Quantum simulation models electromagnetic and structural properties of candidate defence materials.	Advanced indigenous materials for stealth, propulsion and armour. Applies to: Defence materials; aerospace; metallurgy.

14. Education, Skilling & Research

5 use cases

A quantum economy is ultimately a talent economy. The QAIC's statewide hub-and-spoke network across the Amaravati, Visakhapatnam and Rayalaseema regions — anchored by lead institutions for operations, education and research — is itself the engine that makes every other use case achievable, building the workforce, curricula and research base AP needs.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-090	Statewide quantum talent pipeline and curriculum Quantum adoption is constrained by an acute shortage of skilled talent; AP needs a scaled, structured pipeline.	Quantum Computing · Quantum AI / ML · Quantum Communication The QAIC hub-and-spoke network delivers a tiered quantum/AI curriculum across 380+ college nodes, with regional leads coordinating education, operations and research.	Thousands of quantum-literate graduates and a deep talent base anchoring AP's hub ambition; aligns with AICTE quantum curriculum and NQM skilling. Applies to: Higher Education; APSSDC; universities; industry.
UC-091	Quantum learning platform and simulator access Students lack hands-on access to quantum hardware and tools.	Quantum Computing · Quantum AI / ML A cloud platform provides emulator/simulator and metered real-hardware access (via AQV and cloud providers) with guided projects.	Practical, project-based learning that produces job-ready quantum engineers. Applies to: Universities; colleges; skilling programmes.
UC-092	Adaptive, personalised learning with quantum ML Large, diverse student populations need personalised learning paths that classical systems approximate imperfectly.	Quantum AI / ML QML models learner patterns to recommend personalised content and interventions.	Improved learning outcomes and reduced dropout across AP's education system. Applies to: School & higher education; ed-tech; skilling.
UC-093	Quantum cloud access and research collaboration grid Research institutions across AP need shared, governed access to quantum resources and collaboration infrastructure.	Quantum Computing · Quantum AI / ML · Quantum Communication A statewide quantum-simulation and access grid (building on Param Shavak/Rudra-class platforms and AQV hardware) connects colleges to compute and to each other.	Democratised research access and a collaborative quantum research community across the three regions. Applies to: Research institutions; universities; startups.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-094	Faculty upskilling and quantum-ready workforce certification Faculty and professionals need credible, current quantum training to teach and apply the technology.	Quantum Computing · Quantum AI / ML · Quantum Communication Structured faculty-development and industry-certification programmes (with WISER-style women-in-quantum tracks) build teaching and workforce capacity.	A self-sustaining base of educators and certified professionals enabling all other use cases. Applies to: Higher Education; APSSDC; industry HR.

15. Space & Satellite Technologies

6 use cases

Space is both a beneficiary of quantum computing (optimisation, data analysis, materials) and a frontier for deploying quantum technology in orbit (space-based QKD, orbital error correction). With India's strong space ecosystem and AP's space-industry presence (Sriharikota proximity), space-quantum convergence is a strategic priority, pairing data-analysis and optimisation wins with frontier in-orbit quantum technology.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
UC-095	Space-hardened quantum error correction (orbital QC) Deploying quantum computers in orbit faces space-radiation effects requiring novel, space-tailored error correction.	Quantum Computing Space-hardened QEC protocols, radiation-environment modelling and suitable shielding (e.g. high-hydrogen plastics) enable resilient orbital quantum computing; protocols have been demonstrated on real hardware.	Foundations for a quantum-computing constellation supporting secure space communications and processing. Applies to: Space agencies; defence; satellite operators.
UC-096	Satellite constellation and orbit optimisation Designing constellations, scheduling tasking and planning collision-avoidance manoeuvres are complex optimisation problems.	Quantum Computing Quantum optimisation plans constellation geometry, tasking and manoeuvres under coverage and fuel constraints.	Better coverage, longer satellite life and safer operations. Applies to: Space operators; earth-observation; communications.
UC-097	Earth-observation data analysis with quantum ML Satellite imagery volumes overwhelm classical analysis for land-use, crop, water and disaster monitoring.	Quantum AI / ML QML classifies and detects change in high-dimensional satellite imagery more efficiently.	Faster, cheaper insights for agriculture, water, urban and disaster monitoring across AP. Applies to: Agriculture; water; urban planning; disaster management.
UC-098	Space-based quantum key distribution	Quantum Communication Satellite QKD distributes entanglement/keys over continental distances, complementing fibre	Nation-scale and international quantum-secure communication capability. Applies to: Government; defence; finance; international links.

#	Use Case & Problem	Quantum Approach & How It Helps	Expected Impact & Applicability
	Ground QKD is range-limited by fibre loss; global secure communication needs satellite relays.	networks within the NQM quantum-communication programme.	
UC-099	Launch trajectory and mission-planning optimisation Optimising launch windows, trajectories and payload manifests under many constraints is computationally intensive.	Quantum Computing Quantum optimisation explores trajectory and manifest options for fuel and cost efficiency.	More efficient, lower-cost launches and mission plans. Applies to: Space agencies; launch providers; satellite operators.
UC-100	Deep-space materials and propulsion simulation Designing materials and propellants for extreme space conditions requires accurate quantum-scale modelling.	Quantum Computing Quantum simulation models high-temperature, radiation-resistant materials and propulsion chemistry.	Advanced indigenous materials and propulsion for ambitious space missions. Applies to: Space materials; propulsion; defence aerospace.

Sources and References

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